

Project Overview:

This assessment focused on reducing endpoint attack surfaces through operating system hardening, malware execution prevention, browser security configuration, and Linux service management. The project emphasized defensive controls that improve baseline system security.

Tool Used

- Windows Security Features
- Microsoft Edge
- Linux Administration Utilities
- systemctl

Skills Demonstrated

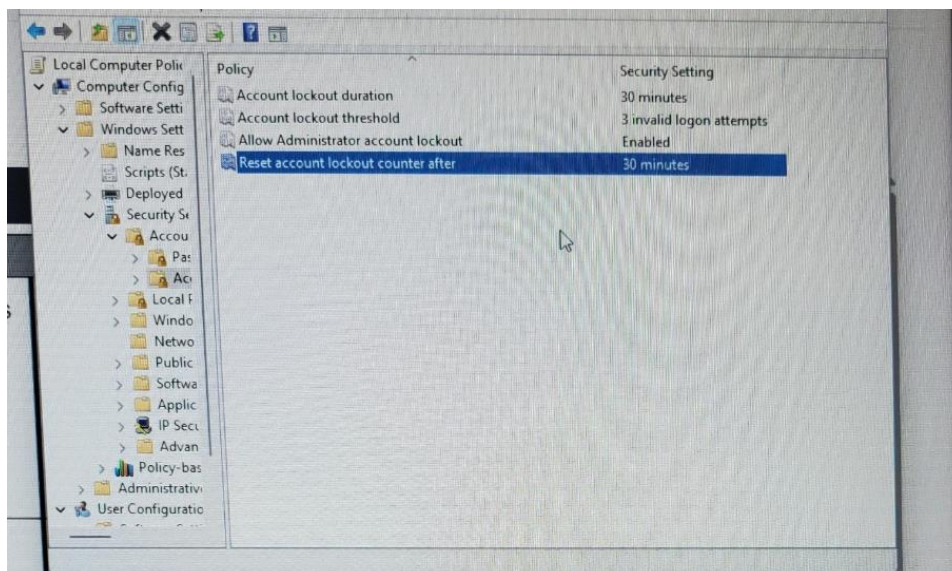
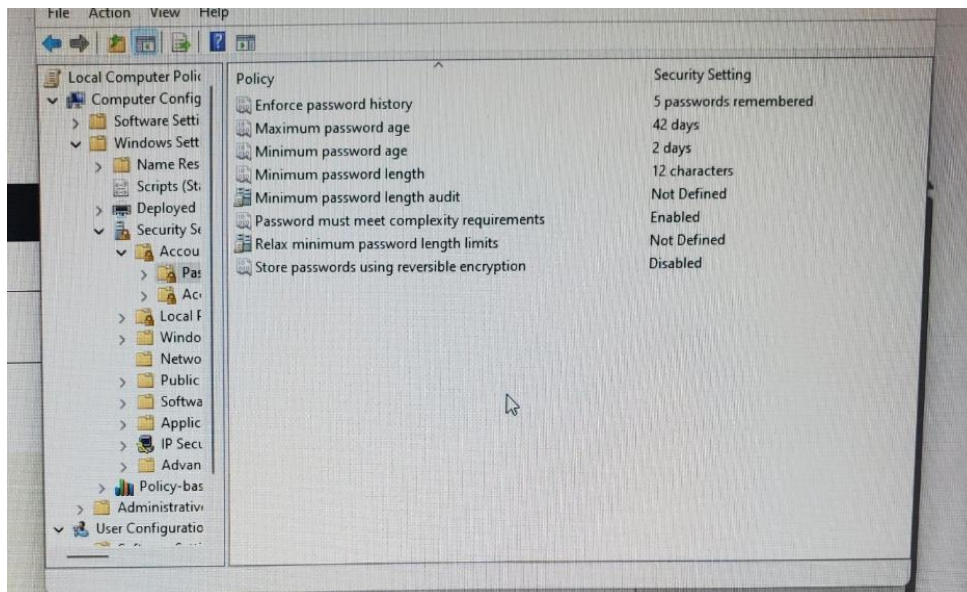
- Endpoint Security
- System Hardening
- Browser Security Configuration
- Linux Administration
- Service Management
- Malware Prevention

Summary:

The assessment demonstrated how secure configuration management can reduce risk without introducing additional security products. DEP was evaluated as a mitigation against memory-based attacks, browser privacy settings were hardened, and unnecessary Linux services were disabled to reduce exposure.

Phase 1. Data Execution Prevention (DEP)

DEP settings were reviewed and configured to ensure executable code cannot run from memory regions intended only for data. DEP adds a meaningful barrier against buffer overflow and arbitrary code execution techniques by enforcing memory protection policies. Disabling DEP increases exposure to exploit classes that depend on executing payloads from writable memory regions.



Phase 2: Browser Security Hardening

Microsoft Edge was configured to block third-party cookies and review privacy-related controls. Blocking third-party cookies reduced cross-site tracking and limited data collection by external parties. Browser hardening helps reduce tracking, unwanted profiling, and some web-based attack exposure.

Phase 3: Process and Service Hardening

System processes were reviewed, and SSH service management was demonstrated in a controlled Linux environment. Disabling services that are not operationally required reduce potential attack vectors and narrows the externally reachable surface.

samba-bgqd.service	disabled	enabled
saned.service	disabled	disabled
saned@.service	masked	disabled
screen-cleanup.service	indirect	disabled
serial-getty@.service	masked	disabled
smartd.service	disabled	disabled
smartmontools.service	alias	-
smbd.service	enabled	enabled
snmpd.service	disabled	disabled
speech-dispatcher.service	disabled	disabled
speech-dispatcherd.service	alias	-
ssh.service	disabled	disabled
sshd-keygen.service	disabled	disabled
sshd-unix-local@.service	disabled	disabled
sshd@.service	alias	-
ssllh.service	indirect	disabled
	disabled	disabled

Lesson Learned:

The project demonstrated that meaningful security improvements often come from configuration management rather than new tooling. Small changes to memory protection, browser privacy settings, and service exposure can materially improve endpoint resilience.

Conclusion:

This assessment provided practical experience with operating system hardening, browser security configuration, malware execution prevention, and Linux service management. The resulting controls reduced attack surface and improved baseline endpoint security posture.